MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEBOT201 - HORTICULTURE AND POST-HARVEST TECHNOLOGY
(2024 ADMISSION ONWARDS)

Answer Key

Duration: 2 hours Maximum Marks: 70

Part A
Short answer
Answer any 10 questions
Each question carries 2 marks

1. Give one example of environmental landscaping in urban areas

   Urban parks / Vertical gardens / Green roofs.

2. What is meant by soil profile (soil horizon)?

   Vertical section of soil showing different layers (horizons).

3. Which tool is specifically designed for planting bulbs?

   Bulb planter or any other suitable tool

4. What is the purpose of winding paths in English gardens?
   To create a natural and aesthetic appearance

5. Which chemical is frequently used as a sanitizer in produce wash water?
   Chlorine (e.g., sodium hypochlorite)

6. What is the typical gas composition used in Controlled Atmospheric (CA)
   Low O (2–5%) and high CO (3–10%)

7. What is meant by green manure?
   Crops grown and incorporated into soil to improve fertility

8. State one advantage of budding over grafting.
   Requires less plant material / faster / higher success rate

9. Name one objective of packaging fresh produce.
   Protection during transport / prolong shelf life

10. Name one government-owned agency promoting horticultural products.
    APEDA / NHB (National Horticulture Board)

11. Which sector provides large employment opportunities in rural areas?
    Agriculture / Horticulture sector

12. Name one quality standard for processed food products.
    FSSAI / AGMARK / ISO

[2x10 = 20]

Part B

Short Essay

Answer any 8 questions

Each question carries 5 marks

13. Describe the characteristics of organic soil.

- High organic matter
- Good water holding capacity
- Dark color
- Rich microbial activity

14. Define sub soil (subsurface) irrigation.
    A method of irrigation in which water is supplied below the soil surface directly to the root zone through buried pipes or channels, maintaining adequate moisture without wetting the surface.
15. Explain the role of floral oils in commercial floriculture.

- Used in perfumes
- Value addition
- Export commodity
- Aromatherapy uses

16. Describe briefly about the branches of horticulture.

- Pomology
- Olericulture
- Floriculture
- Plantation crops
- Landscape gardening

    (Describe each point briefly)

17. What are the characteristics of sodic soil?

- High sodium content
- Poor structure
- Low permeability
- High pH

18. What is Integrated Pest Management?
    An eco-friendly approach to pest control that combines biological, cultural, mechanical, and chemical methods in a coordinated way to manage pest populations

below economic damage levels while minimizing harm to the environment, human health, and beneficial organisms.

19. Explain the role of legumes in vegetable farming?

- Nitrogen fixation
- Improve soil fertility
- Reduce fertilizer requirement

20. Explain the technical requirements for successful Tongue grafting.

- Compatible stock and scion
- Proper alignment of cambium
- Clean cuts
- Proper tying and care

21. Write a short note on business opportunities in food processing sector.

- Value addition
- Employment generation
- Export potential
- Small-scale industries

(Describe each point briefly)

22. Differentiate between jam and jelly.

Feature	Jam	Jelly
Material	Crushed fruit	Fruit juice
Texture	Thick with pulp	Clear and firm
Appearance	Opaque	Transparent

[5x8 = 40]

Part C

Essay

Answer any 1 question

Each question carries 10 marks

23. Provide a detailed comparison of India's three major fruit climatic zones

1. Tropical Zone

- High temperature and heavy rainfall
- No frost; warm throughout the year
- Regions: Kerala, Tamil Nadu, Karnataka, Andhra Pradesh

- Crops: Mango, banana, coconut, papaya, pineapple
- Continuous or multiple fruiting cycles

2. Subtropical Zone

- Moderate climate with distinct summer and winter
- Mild frost may occur
- Regions: Uttar Pradesh, Bihar, Punjab, West Bengal
- Crops: Citrus, guava, litchi, pomegranate
- Seasonal flowering and fruiting

3. Temperate Zone

- Cold climate with snowfall in winter
- Requires chilling period for flowering
- Regions: Himachal Pradesh, Jammu & Kashmir, Uttarakhand
- Crops: Apple, pear, peach, plum, cherry
- Strong seasonal growth cycle

24. Discuss the role of export promotion agencies in enhancing the export of fresh and processed agricultural products from India

- Provide financial support
- Quality certification
- Market linkage
- Export promotion schemes
- Examples: APEDA, MPEDA

[10x1 = 10]